

National University of Computer and Emerging Sciences



Lab Exercise 14 **AL2002-Artificial Intelligence Lab**

Lab Instructor(s)	Ms. Rida Amir
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Department of Computer Science
FAST-NU, Lahore, Pakistan

Exercise

1. Regression

- Generate a synthetic regression dataset using NumPy.

```
import numpy as np
# Set random seed for reproducibility
np.random.seed(42)
# Generate synthetic data
X = 2 * np.random.rand(100, 1)          # 100 samples, 1 feature
y = 4 + 3 * X.flatten() + np.random.randn(100) # Linear relation
with noise
```

- Display your generated data

```
print("Feature values (X):")
print(X[:10])    # show first 10 samples

print("\nTarget values (y):")
print(y[:10])    # show first 10 target values
```

- Split the dataset into training and testing sets.
- Using your knowledge from Lab 9, use scikit-learn to train a Linear Regression model.
- Generate predictions on testset.

(1) Write Python functions (using only NumPy) to calculate the following metrics:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2)

(2) Use Scikit-learn's built-in functions to compute the same metrics.

(3) Display both manual and built-in results side-by-side to verify correctness.

2. Classification

- Load the attached dataset (*M* in diagnosis column refers to malignant-cancerous and *B* refers to benign-not cancerous)

- Split the data into training (80%) and testing (20%) sets.
- Using your knowledge from Lab 9, train any machine learning model from scikit-learn.
- Implement Confusion matrix from scratch
- Using the matrix, find Accuracy, Precision, Recall, F1 from scratch.
- Verify values for these metrics Using Built-in Methods.